

Network establishment of arbuscular mycorrhizal hyphae in the rhizospheres between citrus rootstocks and *Paspalum notatum* or *Vulpia myuros* grown in sand substrate

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Abstract A greenhouse experiment was conducted to examine the favorable effects of sod culture system with bahiagrass (*Paspalum notatum* Flüggé.) and *Vulpia myuros* (L.) C. C. Gmel. intercropped with citrus trees on the establishment of the network of arbuscular mycorrhizal (AM) fungus hyphae in their rhizospheres. Special acrylic root boxes with three compartments were used for the experiment. Four types of citrus rootstock seedlings, trifoliate orange (*Poncirus trifoliata* Raf.), sour orange (*Citrus aurantium* L.), rough lemon (*Citrus jambhiri* Lush.), and *Citrus natsudaoidai* Hayata, were separately transplanted into one outer compartment in each box, and the seedlings of bahiagrass and *V. myuros* were separately transplanted into the other outer compartment. An AM fungus, *Gigaspora margarita* Becker and Hall, was inoculated in the center compartment of each box. Some boxes with both outer compartments without plants and with some plants in only one outer compartment were also prepared. The box with bare × bare had very low density of AM hyphae. There were a few hyphae in bare compartments in the boxes of trifoliate orange × bare, sour orange × bare, rough lemon × bare, and *C. natsudaoidai* × bare. The density of hyphae in the compartments with citrus seedlings and grasses, however, was significantly higher than in every bare compartment, and the hyphae in the compartments with plants penetrated deeply into the sand. In particular, the density in the compartments of citrus seedlings increased when bahiagrass or *V. myuros* was transplanted as a neighboring plant. The percentage of AM

fungus colonization in every plant root was high. New spore formation was observed in compartments with plants, whereas there were few spores in every bare compartment. In particular, the spore formation in bahiagrass compartments was superior to that in other compartments with plants. Our results suggest that the network system by AM hyphae is easily discernible in the rhizospheres between citrus rootstocks and bahiagrass or *V. myuros*, but bare ground severely inhibits the formation and development of AM hyphal network and reduces the number of AM spores in the soil.

Keywords Arbuscular mycorrhizal fungi · Hyphal network · *Paspalum notatum* · Sod culture · *Vulpia myuros*

Introduction

The sod culture system by *Vulpia myuros* is being rapidly adopted in many orchards in Japan due to the benefit of the system. Particularly, the benefits of *V. myuros*-sodded orchards are: 1) no need for herbicides to control weeds, 2) protection against soil erosion, 3) improvement of soil physical properties by roots, 4) increased supply of organic matter to the soil, 5) moderation of acute changes of soil temperature, and 6) increase in soil fauna such as earthworms. The favorable effect of sod culture in orchards where bahiagrass and *V. myuros* are used is attributed to many factors including the stimulation of arbuscular mycorrhizal (AM) fungal activity (Ishii et al. 1996, 2000). Recently, Ishii et al. (2006) reported the existence of some antagonistic bacteria against white root rot fungus, such as *Bacillus subtilis*, *Pseudomonas stutzeri*, *Burkholderia cepacia*, and *Paenibacillus polymyxa*, living outside the shoot and root of *V. myuros* and bahiagrass. These bacteria

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were able to inhibit the growth of other soil-borne phytopathogens, such as *Fusarium oxysporum* and *Pythium ultimum*, and to increase the phosphate solubilization minerals in the culture medium. We are convinced that use of the sod culture system will contribute to a new fruit growing system with lower inputs, sustained fruit productivity, and environmental conservation.

Cruz et al. (2000) reported that the network of AM hyphae was easily observed in the rhizosphere between trifoliate orange and bahiagrass. The hyphal network made by AM fungus interconnections can bring some benefits to the plant–soil system, such as the extension of root longevity (Tommerup and Abbott 1981), and also provide channels to allow nutrient transfer between plants (Martins 1993; Xiaolin and Zhang 1997). Hodge (2003) demonstrated that N capture from organic mate-

rial by *Plantago lanceolata* L. and *Brassica napus* L. was reduced when grown together as compared to the monoculture. However, this reduction was alleviated when added the mycorrhizal inoculum *Glomus mosseae*. The benefit of the nutrient transfer through AM hyphae may be explained by an increased efficiency of nutrient utilization, so that nutrients are maintained in the plant–soil system and not lost to the soil by leaching or adsorption.

The fungus plays a regulatory role in the growth dynamics of the symbiosis between fungi and plants when the plants are under intercropping system (Heinemeyer and Filter 2004). However, there is no report on how sod culture system using *V. myuros* affects the development of AM hyphal network. In this study, we observed the distribution of AM hyphae that could form a network

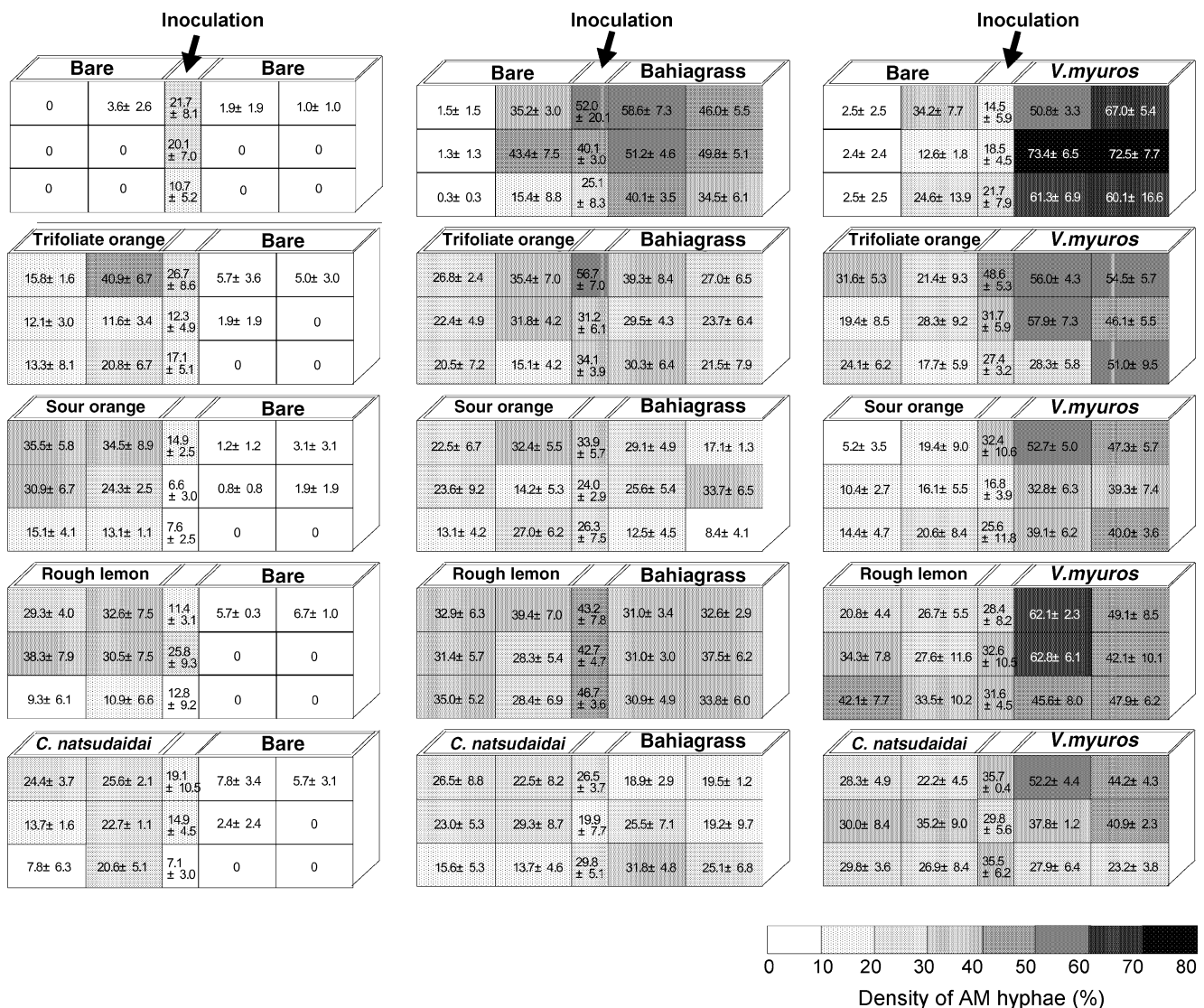


Fig. 1 The distribution of AM hyphae in the root boxes planted with citrus seedlings and grasses. Values of hyphal density in each box indicate mean±standard error (SE; n=3). The distribution of AM hyphae was observed in late July (after 2 months)

system between several kinds of citrus rootstocks and bahiagrass or *V. myuros*.

Materials and methods

This experiment was done in a greenhouse without temperature controls. Special acrylic root boxes were constructed and each root box (3 cm wide, 45 cm long, and 15 cm deep) was divided into three compartments. The center compartment (3 cm wide, 5 cm long and 15 cm deep) was separated from the outer zones by barriers made of a nylon screen of 37- μ m mesh that allows the AM hyphae to penetrate but not the plant roots, thus making easier the observation of the fungal network without root contact. The boxes were covered with aluminum-pasted

film to block the light and prevent the growth of algae. The substrate used was sand, as it is easier to observe the hyphal connections by using CCD camera. Each box was filled with sand sterilized by adding chloropicrin and covering the sand with plastic film. One week later the plastic film was removed and the sand left open for 1 more week to completely release the chloropicrin from the sand.

One citrus rootstock seedling was transplanted into one outer compartment in each box and one seedling of bahiagrass or *V. myuros* was transplanted into the other outer compartment. Citrus rootstocks used were trifoliate orange, sour orange, rough lemon, and *Citrus natsudaoidai*. Boxes without plants were prepared as controls and boxes with one outer compartment was left without a plant. Three boxes in each plot were also prepared. In the center

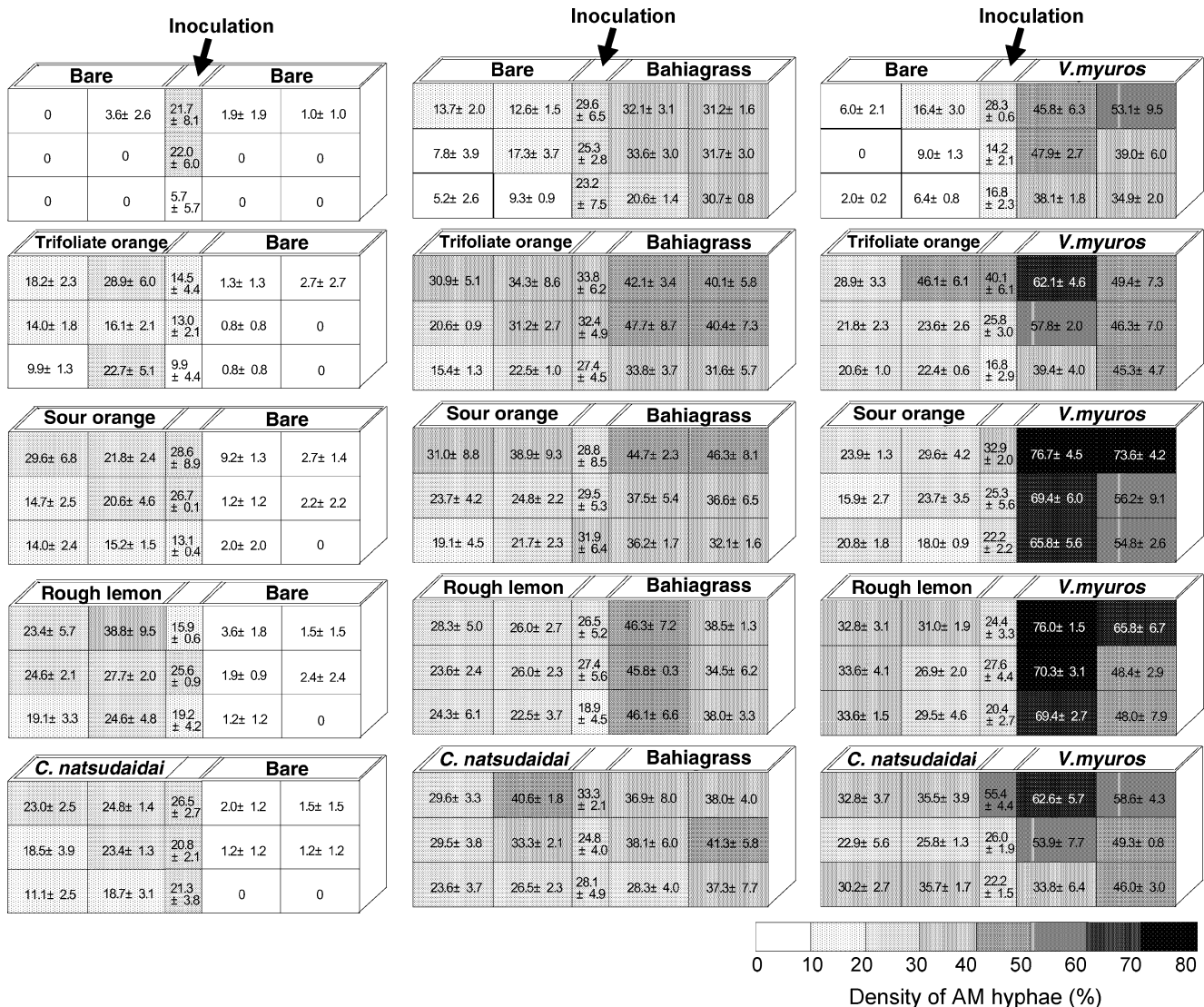


Fig. 2 The distribution of AM hyphae in the root boxes planted with citrus seedlings and grasses. Values of hyphal density in each box indicate mean±SE (n=3). The distribution of AM hyphae was observed in late October (after 5 months)

compartment inoculum containing approximately 100 spores of the AM fungus, *Gigaspora margarita* (Central Glass Co.), was mixed throughout.

Liquid fertilizers containing N, P, K, Ca, and Mg were added to give rates of 300, 180, 300, 120, and 80 mg per box, respectively. In each box 100 ml of a solution containing minor elements such as B, Cu, Fe, Mn, Mo, Co, and Zn were added at once as described by Murashige and Skoog (1962).

In late July and late October (2 and 5 months later), the aluminum cover was removed and the density of hyphae in an area of 12.8×9.3 mm of each compartment was observed using a charge coupled device camera (Keyence VH-7000). On a computer screen, this area was divided into 200 squares and the density of hyphae was calculated by using the following equation: density of AM hyphae (%)=[squares with hyphae/total squares (200)]×100 (Cruz et al. 2000).

After the final observation, plants were harvested and weighed. Root samples were also taken, washed, and stained by the technique of Phillips and Hayman (1970). The percentage of AM fungus colonization in the roots was determined according to Ishii and Kadoya (1994). Sand samples were taken to evaluate the number of spores in 30 g sand according to the procedure of Ishii et al. (1996).

Variance of the means judged statistically from the data obtained was represented as standard error.

Results

Figures 1 and 2 show the distribution of AM hyphae in the root boxes planted with citrus seedlings and grasses, in late July and late October, respectively. There were very few hyphae in the plots of bare × bare controls at any observation time, and hyphae in topsand were observed only in late October (Fig. 2). In the plots of trifoliate orange × bare, sour orange × bare, rough lemon × bare, and *C. natsudaidai* × bare, a slight increase of hyphal density in the bare compartments was observed compared to the controls. In the plots of bahiagrass × bare, and *V. myuros* × bare, however, the density of hyphae in the bare compartments near their grass compartments was higher than that in the plots of trifoliate orange × bare, sour orange × bare, rough lemon × bare, and *C. natsudaidai* × bare. The hyphal density in the bare compartment adjacent to bahiagrass plot in late July was similar to that in late October, but the density in the bare compartment adjacent to *V. myuros* in late October was less than that in late July. The hyphae penetrated deeply into the sand in the boxes with plants on either side. In these boxes, the density of hyphae in the compartments with grasses was higher than those in the compartments with citrus seedlings. Particularly, the hyphal density of *V. myuros* compartments was very high. Furthermore, the hyphal density in compartments with citrus seedlings was greater when paired with a grass than with a bare

Table 1 Growth and AM fungus colonization of citrus seedlings and grasses, and the number of AM spores in the soil after 5 months (late October)

Treatment ^a	Citrus		Grass		AM colonization (%)		AM spores/30 g soil	
	Total FW (g)	Root FW (g)	Total FW (g)	Root FW (g)	Citrus root	Grass root	Citrus or bare soils	Grass or bare soils
No×No	—	—	—	—	—	—	0	0
No×BG	—	—	86.1±3.3	66.3±3.1	—	99.3±0.1	4±0	125±23
No×VM	—	—	14.8±6.0	5.0±3.0	—	93.7±2.7	1±0	3±0
TO×No	6.2±0.3 ^b	3.8±0.4	—	—	83.3±9.5	—	4±1	1±0
TO×BG	6.5±0.4	4.4±0.5	56.4±1.2	33.7±1.1	98.4±0.6	99.9±0.1	4±0	52±16
TO×VM	7.5±0.9	4.7±0.6	14.0±3.2	6.9±2.7	93.6±2.0	95.6±1.0	5±0	12±2
SO×No	12.3±1.8	7.6±0.8	—	—	95.1±1.4	—	6±4	2±0
SO×BG	10.8±0.3	5.5±0.4	57.8±2.5	44.8±1.6	98.4±0.6	99.4±0.3	2±0	93±10
SO×VM	11.7±1.0	5.8±0.5	21.7±2.9	12.2±1.3	94.6±0.6	92.6±1.9	5±1	8±4
RL×No	12.7±0.5	7.3±0.1	—	—	94.3±2.1	—	4±0	1±1
RL×BG	13.1±0.4	7.4±0.2	46.5±4.2	37.6±3.6	98.1±0.9	96.6±0.9	10±2	67±8
RL×VM	12.7±1.8	6.3±1.3	14.3±1.0	6.6±0.6	97.0±0.9	94.3±1.4	4±2	7±2
CN×No	16.3±1.8	9.7±1.4	—	—	89.3±3.5	—	1±0	1±0
CN×BG	14.8±0.6	8.8±0.4	42.1±6.6	33.8±5.0	92.2±0.5	97.5±0.7	3±0	46±15
CN×VM	12.2±1.0	5.8±0.3	12.2±0.5	4.5±0.5	94.7±2.0	96.9±1.1	5±0	13±1

^a No bare (no plants), BG Bahiagrass, VM *V. myuros*, TO trifoliate orange, SO sour orange, RL rough lemon, CN *C. natsudaidai*, FW fresh weight, Total whole plant.

^b Mean±SE (n=3).

compartment. Although the growth of bahiagrass was vigorous in the plots of bahiagrass $\times$ bare, there were no significant differences between treatments on the growth of citrus seedlings in other boxes (Table 1). The percentage of root colonization by AM fungi was very high for all plant species (Table 1). New spore formation was observed in compartments with plants, whereas there were few spores in every bare compartment. In particular, the spore formation in bahiagrass compartments was superior to that with other plants (Table 1).

Discussion

Citrus is considered highly dependent upon AM fungi (Menge et al. 1978; Nemec 1979) because of their short root hairs (approximately 5–100 μ m) (Ishii and Kadoya 1994). The other reason could be the location of these root hairs in patches in the zone 0.3–15 mm behind the root tip under aerobic conditions (Castle and Krezdors 1979; Ishii and Kadoya 1984). By improving the absorption of essential elements, the infected trees grow more rapidly and appear healthier than non-infected trees especially in soils of low fertility (Antunes and Cardoso 1991; Furguson and Menge 1986; Nemec 1979). AM fungus inoculation can also increase the tolerance to water stress (Graham et al. 1987; Shrestha et al. 1996). The photosynthesis and transpiration rates of AM fungus-infected satsuma mandarin trees growing in P-deficient soil were higher than those of non-AM trees when exposed to high temperatures in August (Shrestha et al. 1995). Interestingly, inoculation with AM fungi improved the fruit quality of satsuma mandarin trees, and in particular, it enhanced the Hunter's *a/b* value of peel color and the sugar content in juice (Shrestha et al. 1996).

Our results suggest that increase in number of AM propagules and root colonization of citrus orchards sodded with *V. myuros* and bahiagrass will enhance the formation of the network of AM hyphae. Perhaps root exudates released from bahiagrass or neginatagaya stimulated the fungal micelial formation and thus this network was also formed near citrus roots. Cruz et al. (2000, 2002) also indicated that the introduction of bahiagrass to citrus orchards with trifoliate orange rootstocks would promote the establishment of AM hyphal networks. The trifoliate orange rootstock is most popular in Japan and is used in most satsuma mandarin orchards. In this experiment, we also noted the establishment of well-supplied hyphal networks among other citrus rootstocks and *V. myuros*. Under natural conditions this great network system by AM hyphae between the rhizosphere of these plants may alleviate the competition effects between citrus trees and grasses such as *V. myuros* and bahiagrass for absorption of

nutrients and water. In this experiment, we did not observe the direct interconnection of plants by hyphae.

The present data showed that unplanted compartments had very low hyphal densities and few new spores of *Gi. margarita*. It has been reported that hyphal densities in bare ground may be inferior even to those in soils with non-colonized or slightly colonized plants (Cruz et al. 2002). These results imply that the propagation of AM fungi between crop rows in agriculture may be severely damaged by clean cultivation.

In this experiment, the penetration of hyphae into the bare compartment in the boxes of bahiagrass $\times$ bare and *V. myuros* $\times$ bare was observed, and the density of hyphae in the compartments of citrus seedlings increased when bahiagrass or *V. myuros* was grown in the adjacent compartment. Probably the root exudates from these grasses could stimulate the development of the network by AM hyphae in the sand. Indeed the germination of spores and growth of hyphae may be stimulated by root exudates (Cruz et al. 2000), such as flavonoids (Tsai and Phillips 1991). Cruz et al. (2002) also suggest that some compounds released from bahiagrass and millet (*Pennisetum glaucum* L. R. Br.) roots probably acted as signals for attracting AM hyphae. Also the eupalitin, a flavonoid released by bahiagrass roots, could stimulate AM fungus (Ishii et al. 1997). Recently, we found that short molecular peptides such as tryptophan dimer (Trp–Trp) and Leu–Pro would play an important role in AMF symbiosis (Horii and Ishii 2006). In particular, Trp–Trp was abundantly accumulated in water-stressed bahiagrass roots and exuded from the roots to soil, although it was scarcely detected in non-stressed roots. Interestingly, this peptide strongly attracted the hyphae of *Gigaspora margarita* and *Glomus caledonium* and promoted their hyphal growth. Further studies are now required to clarify how these chemical compounds operate upon AMF network establishment in soil. Further studies are also required to identify the chemical compounds from these plant roots, which can stimulate AM network establishment in soils.

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